

AI Rewards Concierge

Intelligent Personalized Recommendations Using Semantic Search & Vector Embeddings

Executive Summary

The AI Rewards Concierge is an intelligent recommendation system that matches users with personalized reward cards, promotional offers, and cashback coupons based on their shopping habits and preferences. By combining semantic search, vector embeddings, and recommendation algorithms, the system delivers highly relevant suggestions that enhance customer engagement and loyalty program effectiveness.

Problem Statement

Current reward programs suffer from poor relevance and user engagement:

- Users receive generic recommendations irrelevant to their spending patterns
- Reward cards are manually matched to user demographics, not behavioral data
- No semantic understanding of offer descriptions relative to user preferences
- Limited scalability when new offers and user segments are introduced

Proposed Solution

Core Concept

Build a semantic recommendation engine that understands both user preferences and reward card characteristics through natural language processing and vector embeddings. The system transforms user shopping profiles and reward card descriptions into dense vector representations, enabling intelligent similarity matching and personalized ranking.

User Input & Profiling

Users provide their shopping preferences through a simple interface:

- Primary spending categories (dining, travel, groceries, entertainment, etc.)
- Monthly spending amount and preferred merchants
- Special preferences (cashback vs. points, premium benefits, forex benefits, etc.)
- User feedback on previous recommendations (like/dislike)

We also consider user profile, money balance, regional constraints.

Technical Architecture & Algorithm Stack

Phase 1: Baseline - TF-IDF + Cosine Similarity

Initial implementation establishes proof-of-concept:

- Vectorize user preference text and reward card descriptions using TF-IDF
- Compute cosine similarity between user profile vector and all reward card vectors
- Rank cards by similarity score and return top-N recommendations
- Advantages: Fast, interpretable, requires no training

Phase 2: Advanced - BERT Embeddings

Upgrade to contextual embeddings for semantic understanding:

- Use pre-trained BERT model to generate 768-dimensional embeddings for user preferences and card descriptions
- BERT captures semantic meaning: "dining" and "restaurants" now recognized as similar concepts
- Compute cosine similarity between BERT embeddings for ranking
- Advantages: Contextual understanding, handles synonyms, captures intent

Phase 3: Scalable - FAISS Vector Search

Optimize for production-scale retrieval:

- Store all reward card BERT embeddings in FAISS (Facebook AI Similarity Search) index
- User preference embedding queried against FAISS index for nearest neighbors ($O(\log n)$ instead of $O(n)$)
- Enables real-time recommendations with 10,000+ reward cards without latency
- Advantages: Scales to millions of items, sub-millisecond retrieval, minimal memory footprint

Implementation Details

Tech Stack

Component	Technology
Backend	Python (FastAPI), scikit-learn, transformers (Hugging Face)
Embeddings	BERT (bert-base-uncased), TF-IDF vectorizer
Vector Search	FAISS (IndexFlatL2, IndexIVFFlat for scaling)
Frontend	React with card carousel and sorting UI
Database	MongoDB

System Workflow

- User inputs of shopping preferences and monthly spending and his/her profile is analyzed.
- Backend preprocesses and vectorizes user input using BERT embedding model
- FAISS index retrieves top 50 similar reward cards from the vector database
- Re-ranking layer applies business rules: bonus multipliers, exclusion filters, margin optimization
- Frontend displays top 5-10 personalized reward cards with match scores and benefits
- User feedback (like/dislike/apply) is logged for continuous model improvement

Demonstration Scenarios

Scenario 1: Frequent Traveler

Input: User spends 40% on airlines, 30% on hotels, loves international travel

System Output: Prioritizes airline co-branded cards, hotel partnership cards, and premium travel benefits. Semantic understanding links "economy class flights" to "frequent flyer perks" without explicit keyword matching.

Scenario 2: Foodie with Discerning Taste

Input: User spends heavily on fine dining, restaurants, and food delivery; preferences include "culinary experiences" and "premium dining"

System Output: Returns dining-focused cards with partnerships at gourmet restaurants, exclusive dining events, and restaurant aggregator platforms. BERT embeddings recognize "gourmet restaurants" as similar to "fine dining" and "culinary experiences".

Scenario 3: Performance Metrics

Dashboard shows: Recommendation accuracy (user applies recommended card), click-through rate, top 10 recommended cards, embedding similarity heatmap comparing TF-IDF vs. BERT performance

Benefits & Differentiators

Advantage	Impact
Semantic Understanding	Matches synonyms & contextual intent, not keyword-based rules
Scalability	FAISS enables sub-millisecond retrieval for 10K+ reward cards
Real-Time Personalization	Adapts recommendations based on user feedback loops
Business Value	Increases card adoption rate and improves customer loyalty metrics

Project Timeline & Scope

Phase 1: Data Preparation & Baseline

- Curate sample dataset: 100+ reward cards with descriptions; 10+ user profiles
- Implement TF-IDF vectorizer and cosine similarity baseline

Phase 2: Advanced Models

- Integrate Hugging Face BERT model for embedding generation
- Compare TF-IDF vs. BERT performance metrics

Phase 3: FAISS & Scalability

- Build FAISS index and vector search pipeline
- Benchmark: 10K cards retrieval latency < 10ms

Phase 4: Frontend & Demo

- Build React UI with preference form and recommendation carousel
- Deploy live demo with sample users and cards
- Prepare performance dashboards and comparison metrics

Success Metrics & Evaluation

- Mean Reciprocal Rank (MRR): Measure if user's ideal card appears in top 5 recommendations
- Normalized Discounted Cumulative Gain (NDCG): Evaluate ranking quality
- Latency: System response time for querying (target < 100ms end-to-end)
- BERT vs. TF-IDF Improvement: Quantify accuracy gain from advanced embeddings
- Scalability: Demonstrate consistency at 100, 1K, 10K reward cards

Conclusion

The AI Rewards Concierge project balances practical implementation with cutting-edge NLP techniques. It progresses from interpretable TF-IDF baselines to advanced BERT embeddings and production-grade FAISS search, demonstrating mastery of modern machine learning concepts while remaining achievable within a 6-week timeline. The project delivers a live, demo-friendly system that showcases both technical depth and business value, making it ideal for portfolio presentations and technical interviews.